

DISSERTATION PROPOSAL

MSc Artificial Intelligence • University of Essex Online

Candidate	Pavlos Papachristos
Student ID	PP24589
Programme	MSc Artificial Intelligence – University of Essex Online
Title	Economic Crisis Prediction: Early Warning Systems for Banking Crises, Evaluating the Interpretability–Performance Trade-off Under Structural Non-Stationarity
Primary Dataset	Jordà–Schularick–Taylor (JST) Macrohistory Database R6
Text Dataset	BIS Central Bank Speeches 1997–2020 (FinBERT inference complete — 16,622 speeches)
Submission	August 2026

1. Research Overview

Economic crises inflict severe and lasting damage on output, employment, and public finances, yet predicting their onset reliably remains a central unresolved problem in Macrofinancial research. Traditional early-warning systems (EWS) rely on structured indicators — credit growth, leverage ratios, interest rate spreads — that are transparent and familiar to regulators but cannot capture the qualitative signals embedded in central bank communication. Policymakers have long recognised that the cautious and hedged language of central bank governors often reflects emerging financial stress before it appears in any macro ratio, yet this channel has historically been excluded from quantitative EWS frameworks because processing it at scale was not feasible.

The theoretical basis for expecting sentiment to carry predictive content rests on the central bank communication literature. Morris and Shin (2002) show that public signals from credible institutions — such as central banks — disproportionately anchor private expectations, creating coordination effects in financial markets that can amplify or dampen systemic risk independently of underlying fundamentals. Blinder et al. (2008) document that sustained shifts in the tone of central bank communication alter financial conditions through at least two channels: an expectations channel, where forward guidance shapes beliefs about future policy and credit availability, and a risk-appetite channel, where language signals about systemic vulnerabilities influence investor positioning. The implied causal chain is: communication tone → expectations about financial conditions → credit and asset price dynamics → crisis probability. If this chain operates, then lagged central bank sentiment should carry predictive content for crises that is not already absorbed by contemporaneous macro indicators — the hypothesis this study tests empirically.

This dissertation tests that hypothesis by combining structured Macrofinancial data from the Jordà–Schularick–Taylor (JST) Macrohistory Database (Jordà, Schularick and Taylor, 2017) with FinBERT-derived sentiment scores from the BIS Central Bank Speeches corpus, covering 18 advanced economies over 1997 to 2020 (432 country-year observations). Three model specifications will be estimated and compared: a logistic regression baseline (M1), a Random Forest ensemble (M2), and a gradient-boosted hybrid model augmented with FinBERT sentiment (M3). Approximate, post-hoc, and non-causal SHAP-based feature attribution (Lundberg and Lee, 2017) will be applied throughout. As of mid-March 2026, the data pipeline is complete: FinBERT inference on 16,622 BIS speeches has been processed and merged with the JST panel to form the master modelling dataset.

Research Questions and Hypotheses

Main Research Question: Can the integration of LLM-derived sentiment signals from central bank communications measurably improve the statistical accuracy of ML-based economic crisis prediction models, while preserving sufficient interpretability for risk management applications?

SQ1. Does FinBERT-derived central bank sentiment provide statistically significant predictive power for systemic banking crises, over and above standard Macrofinancial indicators from the JST database?

SQ2. What is the quantifiable trade-off between predictive accuracy and model interpretability when progressing from logistic regression through Random Forest to the gradient-boosted hybrid?

SQ3. Which macroeconomic and sentiment features are most predictive in the one-to-two-year pre-crisis window, and do the SHAP-attributed drivers align with the financial vulnerability literature?

Hypotheses

H1 (Discrimination): It is hypothesised that M3 will achieve statistically significantly higher discriminative performance than a macro-only model of equivalent complexity, indicating a genuine predictive contribution from FinBERT sentiment features beyond what Macrofinancial indicators capture alone.

H2 (Feature relevance): It is hypothesised that FinBERT sentiment features will contribute materially to M3's predictions, appearing as substantive drivers of crisis probability in global feature attribution analysis rather than being absorbed by macro co-variates.

H3 (Interpretability trade-off): It is hypothesised that a quantifiable accuracy–interpretability trade-off will be observable across the M1 → M2 → M3 progression, with accuracy gains from non-linearity and sentiment augmentation accompanied by a reduction in native model transparency.

2. Study Period and Data

The 1997–2020 window is determined by data availability. The lower bound reflects the start of the BIS speeches corpus; beginning earlier would introduce a structural break, with no sentiment signal available for pre-1997 observations. The upper bound is set by JST Release 6. The window captures two structurally distinct stress episodes: the 1997 Japanese banking crisis and the Global Financial Crisis of 2007–2009, which affected 13 of the 18 sample countries. The crisis base rate in the study sample is approximately 11.6%, calculated from the available crisis country-year observations against the non-crisis majority. The COVID-19 shock in 2020 is structurally exogenous and will be controlled for separately rather than treated as a standard crisis observation.

Macroeconomic data will be drawn from the JST Macrohistory Database R6, supplemented by IMF data for Australia as an eighteenth economy. Core variables will include credit-to-GDP ratios, short- and long-term interest rates, house price indices, and government debt-to-GDP ratios, alongside the binary crisis indicator. Where the JST crisis chronology and Laeven and Valencia (2020) disagree on crisis timing for a given episode, the more conservative label will be adopted. Textual data are drawn from the BIS Central Bank Speeches corpus, which has already been filtered to the 18 sample economies and processed through the FinBERT pipeline.

3. FinBERT Sentiment Extraction

FinBERT (Huang, Wang and Yang, 2023) is a version of the BERT transformer (Devlin et al., 2019) further pre-trained on financial text and fine-tuned for three-class sentiment classification: positive, negative, and neutral. Its advantage over rule-based tools such as Loughran and McDonald (2011) is contextual

reading — it correctly classifies phrases like 'significant downside risk' as negative even in the absence of surface-level negative vocabulary, a property essential for the characteristically hedged language of central bank communication.

FinBERT inference has been completed on the full BIS corpus of 16,622 speeches, with annual country-level sentiment scores merged with the JST panel to produce the master modelling dataset. Full implementation details, including tokenisation strategy and aggregation procedure, will be documented in the methodology chapter.

As no single public dataset provides pre-computed FinBERT scores for this corpus matched to the 18 JST economies, computing them independently is a methodological contribution of this study. Computed scores will be validated against published transformer-derived central bank communication indices, including Bekaert, Popov and Thaler (2023) and Ferrara and Angino (2022), to provide evidence of construct validity before the modelling stage.

4. Model Architecture and Feature Engineering

Three model specifications will be estimated and compared, progressing from a transparent linear baseline through a non-linear ensemble to a gradient-boosted hybrid.

Model	Feature Set	Interpretability	Role
M1: Logistic Regression	JST macro only	Native log-odds coefficients; fully transparent	Interpretable linear baseline; replicates Schularick and Taylor (2012)
M2: Random Forest	JST macro only	Post-hoc SHAP (TreeExplainer); approximate and non-causal	Non-linear ensemble benchmark; isolates the gain from non-linearity alone
M3: XGBoost Hybrid	JST macro + FinBERT sentiment	Post-hoc SHAP; sentiment features ranked alongside macro features	Primary model; tests whether LLM-derived sentiment adds genuine predictive power over M2

This structure is designed to enable two clean comparisons: M1 to M2 isolates the gain from non-linear modelling with the feature set held constant, while M2 to M3 — or equivalently a macro-only XGBoost against M3 — tests the incremental contribution of FinBERT sentiment at equal model complexity.

The macro feature set will follow the empirical EWS literature, drawing on the variables established by Schularick and Taylor (2012) and Beutel et al. (2019). All features will be lagged one year relative to the crisis indicator to prevent look-ahead bias. FinBERT sentiment variables will be appropriately scaled before inclusion in M3 to ensure comparability of feature attribution across the full feature set.

5. Evaluation Framework

A multi-metric framework will be applied throughout. All metrics will be computed on out-of-sample folds only.

AUROC will serve as the primary discrimination metric, measuring the probability that a crisis country-year receives a higher predicted score than a non-crisis one. Published EWS models on JST-type data typically achieve values of 0.70–0.85 (Beutel et al., 2019; Holopainen and Sarlin, 2017), providing a contextual benchmark for the three specifications. Given the class imbalance in the study sample, bootstrap confidence intervals and a DeLong test will be used to assess whether differences between specifications are statistically meaningful.

PR-AUC will complement AUROC as a more sensitive measure of minority-class performance. Under class imbalance of this magnitude, a no-skill classifier would achieve PR-AUC equal to the base rate, and meaningful models are expected to substantially exceed this floor.

Brier Score will be used to assess probability calibration — the correspondence between predicted crisis probabilities and observed frequencies. This is particularly relevant in a macroprudential context, where supervisory actions are often triggered at specific probability thresholds: a model that discriminates well between crisis and non-crisis years but assigns poorly calibrated probabilities may mislead rather than inform the policy decision.

Policy Usefulness Score $U(\theta)$: The Usefulness Score of Alessi and Detken (2018) will be used to translate model performance into policy-relevant terms, explicitly modelling the asymmetric costs faced by a macroprudential regulator who weights missed crises more heavily than false alarms. Reporting this metric across a range of regulator preference parameters will allow the comparison between M2 and M3 to be evaluated not just statistically but in terms of practical decision-making value.

Expanding-window cross-validation: An expanding-window cross-validation design will be used throughout, beginning with an initial training window of 1997–2004 and extending annually to evaluate one-year-ahead predictions from 2005 onwards. A final hold-out test set will be reserved and evaluated exactly once to provide a clean out-of-sample estimate. Class imbalance will be addressed through inverse-frequency weighting, with the specific weight ratio determined by the crisis base rate in the training data.

6. SHAP Interpretability Analysis

SHAP (Lundberg and Lee, 2017) will be applied to M2 and M3 to provide post-hoc feature attribution. An important qualification governs the interpretation of all SHAP outputs: SHAP explains the model's statistical behaviour, not the underlying causal structure of crises. SHAP values indicate which features the model relied upon to generate a prediction; they do not establish why those features drive crises.

All interpretability claims are accordingly approximate, post-hoc, and non-causal, and will be presented as such throughout the dissertation.

Analysis will operate at two levels. At the global level, feature importance plots will be used to compare the contribution of FinBERT sentiment variables against established macro predictors across M2 and M3, directly addressing the question of whether the LLM component adds substantive predictive value. At the local level, individual country-year attributions will be examined for selected pre-crisis periods, connecting model behaviour to the financial vulnerability literature's account of crisis build-up. The progression from M1 through M2 to M3 will be used to quantify the interpretability–accuracy trade-off that is central to the study's research questions.

7. Robustness Checks

Robustness checks will address four areas.

- First, a macro-only XGBoost model with identical architecture to M3 but excluding sentiment features will be estimated to isolate the contribution of FinBERT variables while holding model complexity constant.
- Second, an alternative FinBERT checkpoint will be applied to the same corpus to assess whether results are specific to the primary model implementation or reflect a more general sentiment signal.
- Third, the treatment of the 2020 pandemic year will be examined, given that COVID-driven negative sentiment may not reflect endogenous financial vulnerability.
- Fourth, M1 and M3 will be re-estimated using the Laeven and Valencia (2020) crisis chronology as an alternative label source to assess sensitivity to crisis dating.

8. Limitations

The fundamental constraint is the scarcity of crisis episodes. With approximately 50 positive observations in a dataset of 432, statistical power for detecting AUROC differences between M2 and M3 is materially limited: DeLong confidence intervals are likely to be wide, meaning that small genuine differences in predictive performance may be indistinguishable from sampling noise. This also increases the variance of SHAP importance estimates across cross-validation folds, making global feature rankings less stable than they would be with a larger crisis sample. The complexity of M3 relative to the available positive class observations heightens the risk of overfitting even with imbalance treatment and expanding-window evaluation. Results should be interpreted as indicative rather than definitive, and generalisability beyond the 18 advanced economies studied is not assumed.

The most serious threat to the internal validity of the sentiment contribution is reverse causality. Central banks routinely adjust their communications in response to deteriorating financial conditions that are already visible in markets — meaning that elevated negative sentiment in year t may simply reflect stress that macro variables will also capture in year t , rather than genuinely anticipating year $t+1$ events. If so, sentiment functions as a concurrent indicator rather than a leading one and its predictive power in a one-year-ahead model would be partly illusory: the model would be picking up lagged reality, not anticipating the future.

Mitigants: The one-year feature lag partially mitigates this concern but does not resolve it, since a central bank reacting to second-half stress in year t still produces sentiment that is structurally prior to year $t+1$ outcomes in calendar terms but not necessarily in informational terms. The analysis will examine whether sentiment features contribute independently of contemporaneous macro variables or primarily as co-movers, and the residual uncertainty will be explicitly disclosed in the results chapter. This limitation cannot be fully resolved without an exogenous shock to central bank communication — a design constraint that is standard in the observational EWS literature.

Annual aggregation smooths over intra-year dynamics that may carry predictive information; a quarterly series would be a natural extension. The sample covers only advanced economies and may not generalise to emerging markets or to crisis types beyond systemic banking failures.

9. Project Timeline

As of mid-March 2026, Notebooks 00–04 are complete and the master dataset is ready. Approximately 15 weeks remain to July 2026 draft submission (target submission date).

Phase	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Key Deliverables
EDA & Feature Engineering																<i>Finalise feature set; class-imbalance diagnostics; temporal plots</i>
Model Training & Evaluation																<i>M1, M2, M3; expanding-window CV; AUROC, PR-AUC, Brier, U(ŷ)</i>
Robustness Checks																<i>Alt. FinBERT checkpoint; COVID dummy; alt. crisis labels</i>
SHAP Analysis & Results																<i>TreeExplainer; beeswarm/waterfall plots; results chapter</i>
Dissertation Write-up																<i>Lit. review, methodology, conclusion; Harvard references</i>
Submission																<i>Turnitin; final formatting; submission by July 2026</i>

10. Ethical Considerations

All data sources — the JST Macrohistory Database, BIS Central Bank Speeches corpus, and Laeven and Valencia (2020) dataset — are publicly available and contain no personal or individual-level data.

The research does not involve human participants and does not require ethics committee review beyond the standard academic declaration required by the University of Essex Online. Model outputs are retrospective and are not intended for live deployment or real-time financial decision-making. FinBERT is used for inference only on publicly available text, consistent with Hugging Face open-source non-commercial academic terms.

References

- Alessi, L. and Detken, C. (2018) 'Identifying excessive credit growth and leverage', *Journal of Financial Stability*, 35, pp. 215–225. doi:10.1016/j.jfs.2017.06.005.
- Bekaert, G., Popov, A. and Thaler, D. (2023) Central bank communication and financial market stress. BIS Working Paper No. 1079. Basel: Bank for International Settlements.
- Beutel, J., List, S. and von Schweinitz, G. (2019) 'Does machine learning help us predict banking crises?', *Journal of Financial Stability*, 45, article 100693. doi:10.1016/j.jfs.2019.07.005.
- Blinder, A.S., Ehrmann, M., Fratzscher, M., de Haan, J. and Jansen, D.J. (2008) 'Central bank communication and monetary policy: A survey of theory and evidence', *Journal of Economic Literature*, 46(4), pp. 910–945. doi:10.1257/jel.46.4.910.
- Devlin, J., Chang, M.W., Lee, K. and Toutanova, K. (2019) 'BERT: Pre-training of deep bidirectional transformers for language understanding', *Proceedings of NAACL-HLT 2019*, pp. 4171–4186.
- Ferrara, L. and Angino, S. (2022) Does clarity make central banks more credible? Assessing ECB communication. ECB Working Paper No. 2671. Frankfurt: European Central Bank.

- Holopainen, M. and Sarlin, P. (2017) 'Toward robust early-warning models: A horse race, ensembles and model uncertainty', *Quantitative Finance*, 17(12), pp. 1933–1963. doi:10.1080/14697688.2017.1357972.
- Huang, A.H., Wang, H. and Yang, Y. (2023) 'FinBERT: A large language model for extracting information from financial text', *Contemporary Accounting Research*, 40(2), pp. 806–841. doi:10.1111/1911-3846.12832.
- Jordà, O., Schularick, M. and Taylor, A.M. (2017) 'Macrofinancial history and the new business cycle facts', *NBER Macroeconomics Annual*, 31(1), pp. 213–263. doi:10.1086/690241.
- Laeven, L. and Valencia, F. (2020) 'Systemic banking crises database II', *IMF Economic Review*, 68(2), pp. 307–361. doi:10.1057/s41308-020-00107-3.
- Loughran, T. and McDonald, B. (2011) 'When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks', *Journal of Finance*, 66(1), pp. 35–65. doi:10.1111/j.1540-6261.2010.01625.x.
- Lundberg, S.M. and Lee, S.I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30, pp. 4765–4774.
- Morris, S. and Shin, H.S. (2002) 'Social value of public information', *American Economic Review*, 92(5), pp. 1521–1534. doi:10.1257/000282802762024610.
- Schularick, M. and Taylor, A.M. (2012) 'Credit booms gone bust: Monetary policy, leverage cycles, and financial crises, 1870–2008', *American Economic Review*, 102(2), pp. 1029–1061. doi:10.1257/aer.102.2.1029.